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Consumer preferences for Mobility-as-a-Service (MaaS) in Australia

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ABSTRACT

Mobility-as-a-Service, or MaaS, offers potential consumers access to multiple transport modes and services, owned and operated by different mobility service providers, through an integrated digital platform for planning, booking and payment. We surveyed 3985 geographically and demographically representative Australians nationwide, to understand consumer demand and willingness to pay for MaaS in Australia. Our analysis reveals that there is definitely a market for MaaS in Australia. Depending on the service offering, we find that up to 46 per cent of the Australian population would be willing to adopt MaaS. On average, consumers prefer pay-as-you-go schemes to bundled schemes that offer unlimited access to one or more transport modes and services at fixed monthly costs. Local public transport, taxis and long-distance public transport are the most popular transport services; bikeshare is the least popular. Willingness to use MaaS is strongly correlated with age and lifecycle stage: young individuals who are employed full-time are most likely to use MaaS; older adults who have retired from the workforce and whose children have left home are least likely to use MaaS. Our analysis identifies Melbourne, Canberra and Sydney as good markets for MaaS trials and early launches.

1. Introduction

Recent advances in transport technologies and services have resulted in the development of the concept of Mobility as a Service (MaaS), where consumers can purchase access to multiple transport modes and services, owned and operated by different mobility service providers, through a single integrated digital platform for planning, booking and payment (Shaheen et al., 2017a; Kamargianni et al., 2016; Sochor et al., 2016; Heikkilä, 2014; Hietanen, 2014). MaaS was first trialed in Gothenburg, Sweden in 2013. Since then, similar services have been introduced in Finland, England, Germany, Austria, France, Italy and Switzerland (for comprehensive reviews of existing MaaS systems worldwide, the reader is referred to, among others, Jittrapirom et al., 2017, and Kamargianni et al., 2016). In Australia, MaaS is in the process of being trialed in the Greater Sydney metropolitan region (iMOVE, 2019); and a number of industry and government actors have shown interest in trialing similar systems in other jurisdictions across the nation (ITS Australia, 2018).

The objectives of this paper are two-fold: (1) to explore consumer preferences in relation to MaaS, with a particular emphasis on the Australian context; and (2) to support the development of suitable MaaS systems for the Australian community. We surveyed 3985 demographically and geographically representative Australians nationwide in 2018. Survey participants were asked about their current travel behaviour; attitudes towards different modes of transport; and preferences for different MaaS systems. This study

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reports some of the key findings that emerged from the survey.

Given the growing interest in MaaS from both the public and private sectors, several studies have examined consumer preferences and expectations around MaaS in recent years (Matyas and Kamargianni, 2019; Ho et al., 2018; Polydoropoulou et al., 2018; Kamargianni et al., 2018; Ratilainen, 2017). The present study contributes to this emerging body of literature in two notable ways.

First, consumer demand for MaaS is likely to vary as a function of what transport modes are offered by a particular MaaS system. Our analysis includes a more comprehensive set of transport modes than previous studies. In particular, we examine consumer demand for MaaS as a function of access to one or more of the following seven transport modes: local public transport, long distance public transport, taxis, car rentals, carshare, rideshare and bikeshare. Most previous studies have neglected to include car rentals, long distance public transport and, to a lesser extent, bikeshare and rideshare services. MaaS is expected to be a major force of disruption for each of these businesses in the same way that shared mobility services have disrupted existing public transport and taxi services (Kamargianni et al., 2016). On one hand, MaaS could divert existing customers towards other modes of transport. On the other, MaaS could help existing transport service providers reach newer markets and expand their customer base. Our analysis reveals that local public transport, taxis and long-distance public transport are the most popular transport services, followed by car rentals and rideshare services. Consumer demand is lowest for carshare and bikeshare services. Findings such as these offer useful insights on the commercial implications of MaaS for each of these businesses.

And second, we have a much larger sample than previous studies: we surveyed 3985 demographically and geographically representative Australians nationwide. Consequently, our analysis is able to offer richer insights on the geographic and demographic determinants of consumer demand. For example, we find that willingness to use MaaS is strongly correlated with age and life cycle stage. Young and middle-aged individuals who are either single or married, with or without children at home, are far more likely to use MaaS. In contrast, older individuals whose children have left home are most unwilling to use MaaS. Similarly, we find that the demand for MaaS is low in rich car-dependent suburban neighbourhoods, high in inner-city neighbourhoods, and highest in low-income suburban neighbourhoods that depend heavily on the public transport network to fulfil their mobility needs. Our analysis is also able to identify multiple niche markets for MaaS, based on current travel behaviour patterns. In particular, individuals with high travel needs, individuals with high rates of motorcycle ownership and individuals with high dependence on mobility devices are very receptive to the concept of MaaS, and show strong willingness to use the service if it were available in the market today.

The remainder of the paper is structured as follows. Section 2 describes the broader context underlying the emergence of MaaS as a concept. Section 3 provides a formal definition for MaaS, and some of the key components underlying the service. Section 4 describes the survey itself, and the survey design process. Section 5 undertakes a descriptive analysis of sample characteristics in terms of different geographic and demographic variables of interest, current travel behaviour patterns, and aggregate preferences for MaaS. Section 6 describes the econometric framework used for the analysis of the stated MaaS purchase behaviour data, and Section 7 presents estimation results from the framework. Section 8 discusses the implications of our findings from the framework for practitioners and policy-makers interested in the provision of MaaS. Finally, Section 9 concludes with a discussion of key findings and directions for future research.

2. Background and context

A number of recent ‘megatrends’ are disrupting the provision of transport services worldwide and reshaping the broader mobility landscape (JPI Urban Europe, 2017; Shaheen et al., 2017a; Corwin et al., 2014). Major advances in information and communication technologies (ICTs) have created a digital economy, where new web-based services, such as e-commerce platforms, video messaging services, digital health services, online distance learning portals, etc., are changing the need and desire to travel (Cohen-Blankshtain and Rotem-Mindali, 2016). Many of these same advances have conspired to result in the emergence of new forms of shared mobility services, as represented by short-term carshare services such as GoGet, rideshare services such as Uber, public bikesharing services such as oBike, and on-demand and demand-responsive transport services, such as Telebus, that are changing how consumers use the transportation system (Shaheen et al., 2017b). The rise of collaborative consumption and the growth in business and consumer interest in shared mobility services reflects a broader transition from an ownership-based economy to an access-based economy (Belk, 2014).

While the emergence of MaaS has been enabled in large part by advances in ICTs, it has been aided by concurrent economic and demographic shifts. The turn of the twenty-first century has seen suggestions of “peak car” (Goodwin and Van Dender, 2013), with stagnant or declining levels of private car use across much of the developed world, including Australia. Studies have ascribed the apparent decline in private car dependence to a combination of economic factors, such as a recessionary global economy and rising oil prices, and demographic factors, such as an ageing population, rising higher education enrollment rates, an increase in the average age of entry into the labor market and the decision to start a family at a later age (see, for example, Vij et al., 2017; McDonald, 2015).

Climate change concerns have contributed to a renewed interest in alternative sustainable modes of transport (Creutzig et al., 2015). The transport sector contributes 17 per cent to greenhouse gas emissions in Australia; private car use for passenger transport constitutes roughly half of the total emissions from the transport sector (DE, 2015). Urbanization has enabled the provision of more sustainable modes of transport, particularly mass public transport services that require high population densities to be economically feasible (Guerra and Cervero, 2011). Cities are currently home to half of the world’s population, and are expected to comprise two-thirds of the world’s population by 2050 (UN DESA, 2014). Together, these factors have heralded a generational shift in terms of both attitudes towards and dependence on a privately owned car as a means of transportation.

It is against this larger backdrop that the concept of MaaS was first proposed in Helsinki, Finland in 2014, following discussions between the Helsinki City Planning Department, the Aalto University School of Engineering, and ITS Finland (Heikkilä, 2014). The

concept has since spread to other parts of the world. For example, the USDOT has developed the related concept of Mobility-on-Demand, “an innovative transportation concept where consumers can access mobility, goods, and services on demand by dispatching or using shared mobility, courier services, unmanned aerial vehicles (UAVs), and public transportation solutions” (Shaheen et al., 2017a). The “as-a-service” model has previously been adopted by other industries with success, including the freight transport sector. For example, third party logistics providers offer integrated operation, warehousing and transportation services to manufacturing firms, with varying degrees of scalability and customization, subject to the firm’s requirements and market conditions. MaaS applies these same service-based principles to the provision of passenger transport, with the objective of offering integrated mobility solutions that are tailored to the needs of individual consumers.

3. MaaS definition and key components

A universally agreed upon definition of MaaS is yet to emerge (Smith et al., 2017). Based on a comprehensive review of academic studies and industry reports, we adopt the following definition:

MaaS systems offer consumers access to multiple transport modes and services, owned and operated by different mobility service providers, through an integrated digital platform for planning, booking and payment.

MaaS providers may offer additional features, such as real-time trip information, travel incentives, etc., but these are not deemed essential. Our definition builds on earlier definitions forwarded by Kamargianni et al. (2016), König et al. (2016), Kamargianni and Matyas (2017), MaaS Alliance (2017) and Shaheen et al. (2017a) that regard integration across different transport operators, payment and ticketing systems, and information and communication technologies as necessary to the provision of MaaS. Over subsequent sections, we describe each of these components in greater detail. In essence, this section provides an update to similar discussions contained in these earlier studies on the definition and scope of MaaS.

3.1. Mobility integration

Central to the concept of MaaS is the creation of a single service provider that offers access to the many different transport services operating within a given geographic area. As Hietanen (2014) writes in his seminal paper that introduced the concept, “The vision is to see the whole transport sector as a co-operative, interconnected eco-system, providing services reflecting the needs of customers. The boundaries between different transport modes are blurred or disappear completely.”

The transport modes included in the service and the size of the geographic area may differ across MaaS providers, often significantly. For example, Whim, a MaaS provider in Helsinki, Finland offers access to urban public transport, car rental, taxi, bike-sharing and e-scooter services operating within the city. At the other end of the spectrum, *Mon Chauffeur*, a MaaS platform built by the French National Railways (SNCF), offers TGV passengers the opportunity to book first or last mile journeys by taxi or private hire vehicles across the country. In this regard, MaaS has been credited with introducing the concept of ‘roaming’ to the provision of mobility services, where MaaS providers offer services to consumers not just in their city of residence, but anywhere in the world (Kamargianni and Matyas, 2017). MaaS systems may or may not include access to mass public transport services, point-to-point transport services and other shared mobility systems. Our definition of MaaS does not view access to a particular mode of transport as essential. Most existing public transport services themselves may be viewed as primitive versions of MaaS, where a single apex governing body typically contracts multiple bus and/or rail operators to provide services in predefined geographic areas.

The provision of MaaS has been motivated primarily in the context of passenger transport, and for the purposes of this study we will be limiting our attention to this context only. It is possible that future MaaS operators could potentially offer freight transport services as well (Shaheen et al., 2017a). For example, Uber has already integrated its urban food delivery service UberEats with its other passenger transport service offerings. Some studies have argued that the integration of passenger and freight transport could be an additional selling point to potential consumers (JPI Urban Europe, 2017). In particular, MaaS systems could help address first and last-mile connectivity problems in the context of goods delivery (König et al., 2016).

Some studies have emphasized the role of customization in the design of MaaS offerings. Hietanen (2014) placed customer needs at the heart of his conceptualization of MaaS. Their vision has been echoed by subsequent studies through the importance placed on “need-based and customized mobility solutions for the users” (Jittrapirom et al., 2017; see also König et al., 2016). As a consequence, most current MaaS providers allow potential customers to selectively purchase access to a subset of transport modes and services on offer, as best meets their mobility needs.

3.2. Ticket and payment integration

An integrated ticketing and payment system is the second central component to MaaS. Most metropolitan regions in Australia and worldwide have multiple public transport service providers with different ticketing and payment systems. Private mobility service providers, offering services such as carshare, rideshare, car rental, taxi and bikeshare, usually have their own independent systems. MaaS aims to make intermodal travel across these different transport modes and service providers as seamless as possible.

MaaS plans may be offered as monthly subscriptions or pay-as-you-go services. The monthly subscription model requires consumers to pay monthly fees for access to a predetermined amount of mobility services, such as unlimited access to urban public transport services, a fixed number of kilometres with a rideshare or taxi service, and a fixed number of hours with a carshare or car rental service. Examples include Whim in Helsinki, Finland and Mobil-Flat in Augsburg, Germany. The pay-as-you-go model bills consumers periodically based on actual usage of different mobility services. Examples include Zipster in Singapore and Renfe-as-a-

Service in Spain.

An integrated ticketing and payment system across different mobility services can lead to improvements in both the cost and convenience of transfers between services (for a recent review of the benefits of integration, the reader is referred to [Chowdhury and Ceder, 2016](#)). However, its impacts on patronage are unclear. Some studies have found that integrated ticketing and payment systems for public transport services can increase public transport patronage by 2 to 5 per cent in the short term (see, for example, [Abrate et al., 2009](#); [Matas, 2004](#)), and as much as 25 per cent in some cases ([Sharaby and Shiftan, 2012](#)). Others are not as enthusiastic. For example, the Scottish Transport Research Planning Group reports, “no conclusive evidence was found that integrated ticketing leads directly to patronage or revenue increases, partly because integrated schemes have apparently not been studied or introduced in isolation” ([SESr, 2004](#)).

In the past, ticket and payment integration has been enforced through smart card technology. Public transport smart cards, such as the Opal card in Sydney, the go card in South East Queensland and the myki in Melbourne, that offer access to public transport services operating in their respective metropolitan regions, have served as prototypes for smart cards that are being used currently by some MaaS providers, such as Zipster in Singapore. However, as public transport operators shift away from paper tickets and smartcards to open-loop systems using contactless cards or mobile apps (e.g. [Hendry, 2020](#)), MaaS providers too are likely to adopt these digital technologies to facilitate payment integration. Integration could extend to other transport modes and service providers. For example, the ADEPT (Automatic Debiting And Electronic Payment For Transport) II project in Thessaloniki, Greece, offered road users electronic cards for the payment of road tolls, parking and public transport ([Blythe 2004](#)). Integration with dynamic travel demand management schemes, such as road congestion charge or incentives for off-peak travel, could provide a useful tool to manage congestion if consumer expectations can be moderated appropriately.

3.3. Information and communication technology (ICT) integration

The integration of service-related information across different transport modes and mobility service providers through a single digital platform is the third essential component to MaaS. In particular, there is an unmet demand for an integrated platform that offers journey planning, booking and real-time information services that are personalized to the needs and behaviours of each consumer ([Chorus et al., 2007](#); [Grotenhuis et al., 2007](#)). Some studies have argued for the inclusion of additional information services as well, such as weather forecasts, synchronization with personal activity calendar, travel history report, etc. ([Jittrapirom et al., 2017](#)). For a recent review of journey planners and related services, the reader is referred to [Esztergár-Kiss and Csizsár \(2015\)](#).

Journey planners allow consumers to visualize, compare and select different modes of travel for a given journey. Journey planners may be multimodal, i.e. recommended routes for selected trips may involve transfers between different modes of transport. Evaluations of existing multimodal journey planners find their benefits to lie in their ability to reduce the time and effort required to collect the relevant information, and to decrease the uncertainty associated with the information thus collected. For example, [Zografos et al. \(2012\)](#) surveyed 425 users of WISETRIP, an international multimodal journey planner, and found that 40 per cent of the sample was willing to pay at least €0.80 every time they used the service. However, their findings are based on a subset of study participants that already use the service. They have not been corroborated by other studies, and should be taken with caution.

Journey planners may be dynamic, i.e. they offer real-time updates based on traffic incidents, network delays, current location, etc. The impacts of real-time information on public transport use have been well studied. Benefits include reduced perceptions of waiting times, increased ease-of-use, better travel time utilization, and greater consumer satisfaction. Some studies have even reported modest but statistically significant increases in public transport patronage of 1 to 2 per cent (see, for example, [Brakewood et al., 2015](#); [Tang and Thakuriah, 2012](#)). However, studies differ on consumers' willingness to pay for access to real-time information. In their review of previous research that has examined the question, [Dziekan and Kottenhoff \(2007\)](#) conclude, “the value of real-time information systems at stops and stations seems to lie in an interval between 5 and 20% of the ticket price for the trip.” Others have argued that consumers are unwilling to pay for information that they expect should be provided by the public transport service provider at no cost ([Neuherz et al., 2000](#)). The same sentiment may apply to the provision of similar information by MaaS providers.

Most journey planners offer tools for personalization based on past histories and indicated preferences, such as favourite destinations, maximal walking distances, familiarity with the local transport system, preferences for time and cost, etc. (see, for example, [Jakob et al., 2014](#); [Spitadakis and Fostieri, 2012](#)). Based on a review of past studies and focus group interviews, [Stopka \(2014\)](#) finds that most consumers expect recommendations from journey planners to be automatically personalized to both the individual and the trip context. For example, consumer preferences may vary, depending on whether they are seeking recommendations for their morning commute or a recreational trip during the weekend. Personalization and contextualization can increase the value of these services to consumers.

4. Survey instrument

To understand consumer preferences and expectations for MaaS, we surveyed 3985 demographically and geographically representative Australians nationwide. The consumer survey was designed following extensive consultations with project stakeholders from Transport for New South Wales, Transport for Victoria, Queensland Department of Transport and Main Roads, Department of Transport in Western Australia, Royal Automobile Association of South Australia, MaaS Australia, Keolis Downer, and Transdev Australasia. The survey was informed by insights that emerged from the interviews with industry and government experts. The survey was reviewed and approved by experts at the Bureau of Infrastructure, Transport and Regional Economics. The final instrument comprised five sections:

A. Current travel behaviour: Respondents were asked about their car and motorcycle ownership; frequency of use of different transport modes; dependence on mobility devices; and household monthly travel expenditure.

B. Preferences for on-demand transport (ODT) services: Respondents were asked about their awareness of and familiarity with ODT. Each respondent was presented four different scenarios with hypothetical ODT services. Respondents were asked to imagine that they have access to the hypothetical ODT service described in the scenario, and asked to indicate how frequently they would use such a service, and for what kinds of trips. We exclude findings from this section of the survey, as it is not directly related to MaaS and not relevant to the present study's objectives¹.

C. Preferences for MaaS schemes: Respondents were asked about their awareness of and familiarity with MaaS. They were then provided with a formal definition of MaaS, to help familiarize them with the concept. Each respondent was subsequently presented four different scenarios, such as the one shown in Fig. 1. For each scenario, respondents were presented two hypothetical MaaS schemes that differ from each other in terms of the transport services that they offer access to, level of ticketing and booking integration, degree of personalization, availability of real-time information, subscription model, and price. The attributes were varied systematically across scenarios and respondents, and could take any of the values listed in Table 1.

To make scenarios as realistic as possible, price was determined as a function of the subscription model and the transport services included with the scheme. Pay-as-you-go schemes had nominal monthly subscription rates, which were systematically varied across scenarios and respondents between \$0, \$5 and \$10 per month, for pay-as-you-go access to one or more transport modes. The monthly subscription costs are for access to features such as ticketing, booking and payment integration, access to real-time information, and personalisation. Any costs associated with actual transport mode use would be extra, and this was explained to survey respondents.

Prepaid schemes offer unlimited access to one or more transport modes, but for significantly higher costs. Price points for different schemes were determined based on the priority ordering of transport modes shown in Table 2, which was drawn up in close consultation with transport service providers and policy-makers who were interviewed as part of the study. For example, a scheme that offers unlimited access only to bikeshare services was priced between \$10 to \$30 per month. At the other end of the spectrum, a scheme that offers unlimited access to long distance public transport (i.e. buses and trains to regional and remote areas) was priced between \$500 and \$1000 per month, regardless of whether or not the scheme offers access to additional transport modes (since all other modes are lower priority). While these prices might appear high, they are comparable to similar MaaS schemes that are commercially available in Europe, such as Whim, which offers unlimited access to taxi, carshare and rideshare services for a monthly cost of €499.

The MaaS scheme attributes were varied systematically across scenarios and respondents according to an orthogonal main effects experimental design plan (OMEP). The OMEP takes an orthogonal fraction of a $(2^{12}3^1)^2 = 150,994,944$ factorial design, as there are two alternatives in a choice set, each alternative with 12 two-level attributes and one three-level attribute. The design plan generated 81 choice sets, eight of which were removed from consideration because both alternatives were identical for all attribute levels. Four sets were randomly allocated to each respondent and counterbalancing was used to control for ordering and blocking effects. For each choice set, respondents were asked to indicate which scheme they prefer, if they would purchase the preferred scheme if it were available in the market today, and for what kinds of trips would they use the scheme.

D. Attitudes: Respondents were asked to state their level of agreement or disagreement with statements measuring their attitudes towards driving, car ownership, public transport, carsharing, ODT, MaaS, and new technologies and services in general.

E. Demographics: Respondents were asked about their age, gender, education, employment, place of residence, household size and structure, and income.

The survey concluded with an open text question to elicit any feedback from respondents about the survey itself. Respondent feedback was largely positive, and specific comments indicated a high level of engagement.

5. Sample characteristics

We recruited 3985 Australians nationwide through an online consumer panel maintained by the market research company Survey Sampling International in March 2018 to give their responses to the web-based survey described in the previous section. Participation was incentivized based on survey completion. Consequently, each of the 3985 individuals in our sample answered all questions in the survey, and there is no missing information (we did allow respondents the option to not disclose their income, as that is a potentially sensitive question). Of the individuals that started the survey, 87 per cent completed the survey.

Sample characteristics are summarised in Table 3, and contrasted against the characteristics of the national population as reported by the 2016 Census. Survey respondents were drawn from New South Wales, Victoria, Queensland, South Australia, Western Australia and the Australian Capital Territory, in rough proportion to the population of these states and territories. We did not recruit respondents from Tasmania and Northern Territory because we did not have project funding partners from these states. However, as

¹ ODT refers to adaptive transport services that use a fleet of vehicles to provide shared flexible transport to consumers, when and where they need it. Many actors in industry and government view ODT services as stepping stones to the development of full service MaaS systems (ITS Australia, 2018). In particular, both ODT and MaaS aim to remove some of the pain points currently experienced by public transport customers through the use of new information and communication technologies. However, they should not be confused with each other. MaaS systems may or may not include access to these additional ODT services, just as they may or may not include access to mass public transport services or point-to-point transport services. MaaS, as it has been broadly defined by the academic and grey literature, does not view access to a particular mode of transport as essential.

Scenario 1 of 4

Imagine that the following Mobility-as-a-Service (MaaS) schemes are available in your state. For each transport service type, imagine that the schemes offer equal access to all operators in your state that provide that service type. For example, a scheme that offers access to car rental services offers access to ALL car rental companies operating in your state.

		Scheme A	Scheme B
Transport Services	Local public transport (buses, trams, local trains, ferries, etc.)	✗	✓
	Long distance buses and trains	✓	✓
	Taxis	✗	✗
	Car rental (e.g. Hertz, Thrifty)	✓	✗
	Carshare (e.g. GoGet)	✗	✗
	Rideshare (e.g. UberX)	✓	✗
	Bikeshare (e.g. Reddygo, Melbourne Bike Share, City Cycles)	✗	✓
Real-time information (e.g. service interruptions, schedule delays)		Available	Available
Personalization (e.g. save preferred services, show wheelchair friendly routes)		Available	Available
Ticketing integration (i.e. can the same card or device be used for all transport services in the Scheme?)		Separate tickets/cards are needed to access different services	Single card and/or device to access all services
Booking integration (i.e. can the same platform be used to book all transport services in the Scheme?)		Bookings when needed must be made with the service provider	Bookings when needed can be made through a single digital platform
Costs		\$500 per month for unlimited access to all services	No monthly cost for pay-as-you-go access to all services

Which of these service schemes do you prefer more?

☐ Scheme A

☐ Scheme B

Would you purchase this scheme, if it was available in the market today?

☐ Yes

☐ No

What kind of trips would you use this scheme for? Please select all that apply. If you wouldn't use this scheme, you can skip this question.

☐ To get to place of employment or education

☐ To get to a friend or family member's place of residence

☐ To run errands, like shop for groceries, bank visits, see the doctor, etc.

☐ For social trips, like eating out, watching a movie at a theatre, visiting a bar, etc.

Fig. 1. Example screenshot of hypothetical scenario to elicit consumer preferences for different MaaS schemes.

the population of these two regions is less than 3 per cent of the national population, this is not a major limitation to our analysis. 80 per cent of the sample was drawn from major metropolitan areas, and the remaining 20 per cent was drawn from regional and remote areas, based on the 2016 Australian Statistical Geography Standard (ASGS) classification. All survey respondents were 18 years or older, with a good spread over all age groups, including older adults (18 per cent of the sample is 65 years and older). In terms of other demographic characteristics, such as gender, education, employment, household size and structure, and income, the sample is roughly representative of the national population. However, compared to the national population, our sample is more urban, more likely to be unemployed, and more educated.

Any differences between the sample population and the national population have been controlled for in our analysis through reweighting. Iterative proportional fitting was used to impute the joint probability distributions across our sample and the national population for different categories of the following demographic variables: state of residence, ASGS classification, gender, age, education level, employment status, household size, household structure, and household income. For each individual in our sample, the weighting factor was calculated by taking the ratio of the probability of observing the individual's demographic characteristics in

Table 1
Attributes and levels for the MaaS stated preference experiments.

Attribute	Level
Local public transport	Available
	Unavailable
Long distance public transport	Available
	Unavailable
Taxis	Available
	Unavailable
Car rentals	Available
	Unavailable
Carshare	Available
	Unavailable
Rideshare	Available
	Unavailable
Bikeshare	Available
	Unavailable
Ticketing integration	Single card and/or device to access all services
	Separate tickets/cards are needed to access different services
Booking integration	Bookings when needed can be made through a single digital platform
	Bookings when needed must be made with the service provider
Real-time information	Available
	Unavailable
Personalization	Available
	Unavailable
Subscription model	Pay-as-you-go access to transport modes
	Prepaid unlimited access to transport modes
Monthly subscription cost	Low
	Medium
	High

Table 2
Payment structure for hypothetical MaaS schemes shown to survey respondents.

Subscription type	Transport modes included	Monthly subscription costs
Pay-as-you-go access	Any modes	\$0, \$5, \$10
Prepaid unlimited access	Bikeshare	\$10, \$20, \$30
	Local public transport	\$50, \$150, \$300
	Taxis	\$300, \$400, \$500
	Rideshare	
	Carshare	
	Car rental	
	Long distance public transport	\$500, \$700, \$1000

Table 3
Summary of sample characteristics and how they compare with the national population.

Variable	Category	Sample	2016 Census
Area of residence	Major cities	80.5%	71.2%
	Regional areas	18.7%	26.7%
	Remote areas	0.8%	2.0%
Gender	Male	48.6%	50.1%
	Female	51.2%	49.9%
Age	18–39 years	38.0%	40.0%
	40–64 years	43.6%	40.5%
	65 years and older	18.4%	19.6%
Employment status	Employed	59.0%	56.1%
	Unemployed	9.6%	4.1%
	Not in the labour force	31.4%	39.8%
Highest level of education	Bachelor's or higher	39.8%	24.7%
	Diploma or certificates I-IV	29.8%	32.8%
	Year 12 or below	30.3%	42.5%
Average household size		2.6	2.5
Average household income		\$1,550 per week	\$1,736 per week

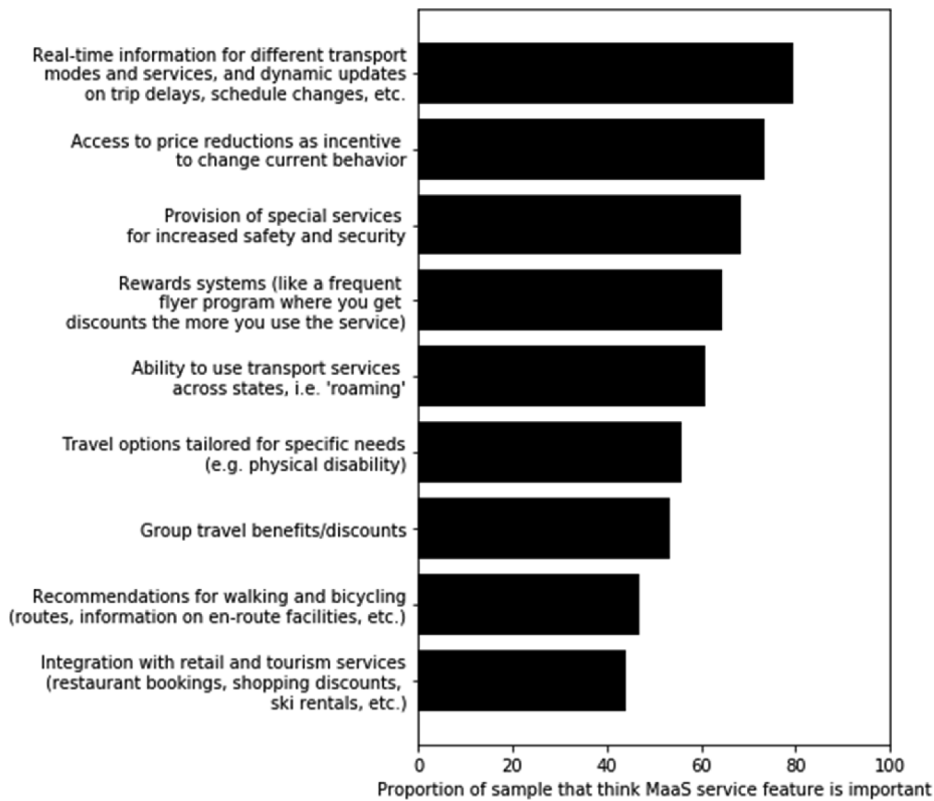


Fig. 2. Ranked importance of different MaaS service features.

the Australian population to the corresponding probability for our sample.

In terms of familiarity with MaaS, only 4 per cent of our sample indicated having heard of the concept of MaaS. Of these, half indicated being only slightly familiar with the concept, and the remainder indicated being moderately or very familiar. As there are no commercial MaaS services currently available in Australia, and the concept is still relatively new globally, low levels of awareness and familiarity are to be expected.

For each of the MaaS scenarios, respondents were asked to indicate what kinds of trips, if any, would they make using their preferred MaaS alternative. MaaS use was reported to be greatest for one-off social trips, like eating out, watching a movie, visiting a bar, etc., with respondents indicating for roughly 40 per cent of the scenarios that they would use their preferred MaaS service to make such trips, indicating that MaaS could help plug service gaps that make these trips relatively inconvenient to make using existing public transport services. Respondents indicated they would use their preferred MaaS alternative for other trips, such as to commute to a place of employment or education, to visit their friends and family members, and to run errands, for roughly 20 per cent of the scenarios.

After the scenarios, we asked survey respondents to indicate the importance of different potential MaaS service features. Fig. 2 ranks these features in order of their average importance. Access to real-time information and dynamic updates, incentives to change travel behaviour, and special services for increased safety and security were ranked as the three most important attributes, serving to emphasize what aspects of service provision are most valued by potential consumers. A number of government experts that we interviewed for the study mentioned how MaaS could potentially be used to implement travel demand management strategies, such as real-time route optimization and dynamic road pricing. From a consumer standpoint, “Access to price reductions as incentive to change behaviour” was rated the second most important feature, indicating that consumers would be receptive to these travel demand management strategies being woven into MaaS systems.

We asked respondents if they had any preferences for how MaaS is delivered in Australia (see Fig. 3). Of our sample, 37 per cent indicated that they had no preference, and this was by far the most popular option. Of those that did indicate a particular preference, there was a strong inclination towards the government playing a role in any potential service provision model, especially as an overseer, but also possibly as a service provider. Roughly 24 and 22 per cent of the sample indicated they would prefer to see MaaS provided by either a private company or a community organization, respectively, with government oversight. 20 per cent of the sample indicated they would prefer to see MaaS provided by the government directly.

Finally, respondents were also asked to indicate their overall attitudes towards MaaS, and its potential impact on car ownership. Fig. 4 plots average attitudes across the sample; many of these statements have been adapted from Kamargianni et al. (2018). On average, respondents are slightly sceptical about the ability of MaaS to impact their decisions related to car ownership. However, as

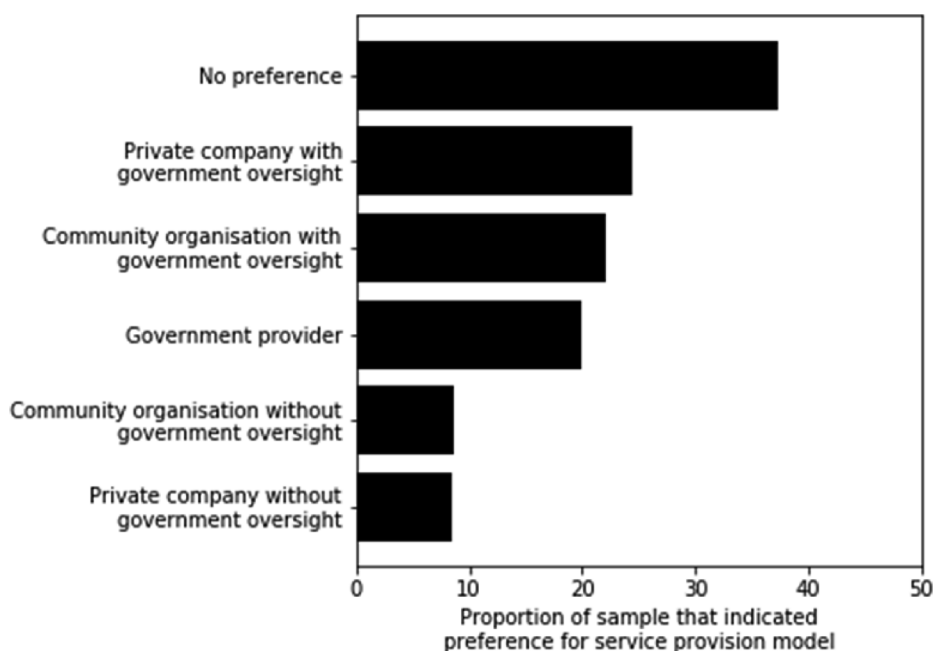


Fig. 3. Consumer preferences for different MaaS provision and governance models.

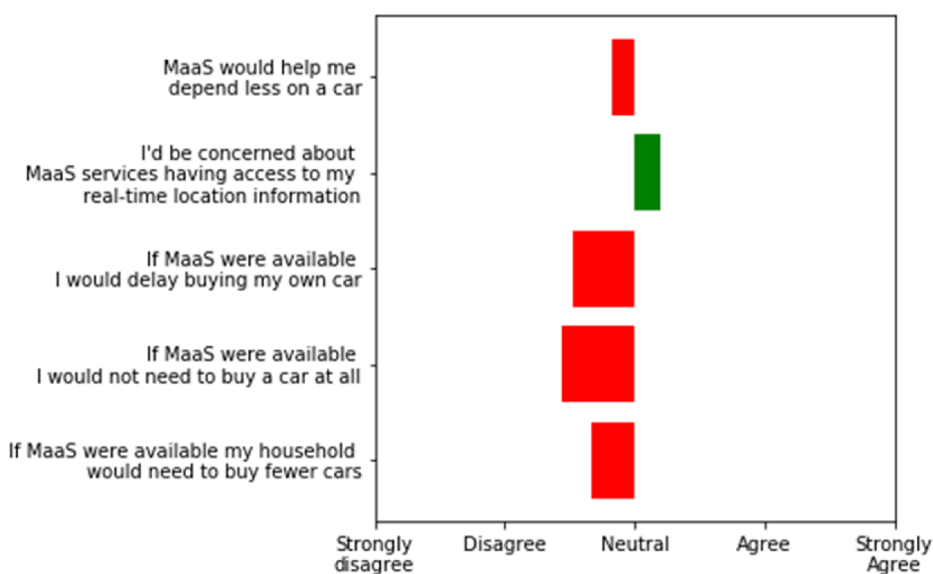


Fig. 4. Average consumer attitudes towards MaaS.

mentioned previously, most respondents in our sample have little to no real-life experience with MaaS services, and it is possible that their attitudes might shift over time as MaaS becomes more widely available in Australia, as has been the case with other innovations (e.g. Rogers, 2010).

6. Econometric framework

Data from the hypothetical scenarios, such as the one shown in Fig. 1, was used in conjunction with other geographic and demographic information collected as part of the survey to estimate latent class choice models (LCCMs) of consumer preferences for MaaS. LCCMs are finite mixtures of discrete choice models. They were first developed in the field of marketing sciences as tools to identify relatively homogenous consumer segments that differ substantially from each other in terms of their behavior in the marketplace (Kamakura and Russell, 1989). They have since emerged as a very popular form of discrete choice model, finding application in a wide variety of disciplines, including but not limited to transportation. In our case, LCCMs allow us to identify segments in the

population that differ in terms of their preferences for different MaaS schemes.

LCCMs comprise two components: a class membership model and a class-specific choice model. The class membership model formulates the probability that a decision-maker belongs to a particular segment, or class, as some function of the characteristics of the decision-maker. Conditioned on the class that the decision-maker belongs to, the class-specific choice model formulates the probability that the decision-maker chooses a particular alternative as some function of the attributes of all of the alternatives in the choice set.

We begin with a description of the class membership model, formulated in our case as the familiar multinomial logit function:

$$P(q_{ns} = 1) = \frac{\exp(\mathbf{z}_n \gamma'_s)}{\sum_{s=1}^S \exp(\mathbf{z}_n \gamma'_s)} \quad (1)$$

where q_{ns} equals one if decision-maker n belongs to class s , and zero otherwise; $\mathbf{z}_n$ is a vector of decision-maker characteristics, such as age, employment and household income; γ_s is a vector of class-specific parameters denoting sensitivity to the decision-maker characteristics; and S is the total number of classes.

Next, we describe the class-specific choice model. Recall that each decision-maker is shown four different scenarios, where each scenario presents a choice between two hypothetical MaaS schemes. The decision-maker is asked to indicate which scheme they prefer, and if they would purchase the preferred scheme if it were available in the market today. Therefore, for a given decision-maker n and scenario t , the class-specific choice model predicts the probability that scheme j is preferred, and that the decision-maker would purchase scheme j if it were available in the market today.

Let $u_{ntj|s}$ be the utility of scheme j for scenario t and decision-maker n , conditional on the decision-maker belonging to class s , specified as follows:

$$u_{ntj|s} = \mathbf{x}_{ntj} \beta'_s + \varepsilon_{ntj|s} \quad (2)$$

where $\mathbf{x}_{ntj}$ is a vector of scheme-specific attributes, such as the availability of particular transport modes, levels of integration, and subscription model; β_s is the vector of class-specific parameters denoting sensitivities to these attributes; and $\varepsilon_{ntj|s}$ is the stochastic component of the utility specification, assumed for the sake of mathematical convenience to be i.i.d. Gumbel with location zero and scale one across schemes, scenarios and decision-makers. Assuming the decision-makers are utility-maximizers, the class-specific probability that scheme j is preferred over the other scheme is given by the logit expression:

$$P(y_{ntj} = 1 | q_{ns} = 1) = P(u_{ntj|s} \geq u_{ntj'|s} \forall j' = 1, \dots, J) = \frac{\exp(\mathbf{x}_{ntj} \beta'_s)}{\sum_{j'=1}^J \exp(\mathbf{x}_{ntj'} \beta'_s)} \quad (3)$$

where y_{ntj} equals one if scheme j is preferred, and zero otherwise; and J is the number of alternatives shown to the decision-maker for any scenario, equal to two in our case. Similarly, let $v_{nt|s}$ be the utility of not purchasing access to either MaaS scheme for scenario t and decision-maker n , conditional on the decision-maker belonging to class s , specified as follows:

$$v_{nt|s} = \alpha_s + \eta_{nt|s} \quad (4)$$

where α_s implicitly denotes the mean class-specific utility of not purchasing a MaaS scheme; and $\eta_{nt|s}$ is the stochastic component, assumed to be i.i.d. Gumbel with location zero and scale one across scenarios and decision-makers. Conditional on scheme j being preferred, the class-specific probability that scheme j is or isn't purchased may then be given by equations (5) and (6), respectively:

$$P(w_{ntj} = 1 | y_{ntj} = 1, q_{ns} = 1) = P(u_{ntj|s} \geq v_{nt|s}) = \frac{\exp(\mathbf{x}_{ntj} \beta'_s)}{\exp(\alpha_s) + \exp(\mathbf{x}_{ntj} \beta'_s)} \quad (5)$$

$$P(w_{ntj} = 0 | y_{ntj} = 1, q_{ns} = 1) = P(u_{ntj|s} < v_{nt|s}) = \frac{\exp(\alpha_s)}{\exp(\alpha_s) + \exp(\mathbf{x}_{ntj} \beta'_s)} \quad (6)$$

where w_{ntj} equals one if scheme j is purchased, and zero otherwise. Equations (5) and (6) may be combined to yield the following class-specific probability of observing w_{ntj} :

$$P(w_{ntj} | y_{ntj} = 1, q_{ns} = 1) = [P(w_{ntj} = 1 | y_{ntj} = 1, q_{ns} = 1)]^{w_{ntj}} [P(w_{ntj} = 0 | y_{ntj} = 1, q_{ns} = 1)]^{1-w_{ntj}} \quad (7)$$

The reader should note that heterogeneity in the decision-making process is captured by allowing the taste parameters α_s and β_s to vary across classes. Equations (3) and (7) may be combined iteratively over schemes and scenarios to yield the following class-specific probability of observing the vectors of choices $\mathbf{y}_n$ and $\mathbf{w}_n$:

$$P(\mathbf{y}_n, \mathbf{w}_n | q_{ns} = 1) = \prod_{t=1}^T \prod_{j=1}^J [P(y_{ntj} = 1 | q_{ns} = 1) P(w_{ntj} | y_{ntj} = 1, q_{ns} = 1)]^{y_{ntj}} \quad (8)$$

where $\mathbf{y}_n = \langle y_{n11}, \dots, y_{nTJ} \rangle$ and $\mathbf{w}_n = \langle w_{n11}, \dots, w_{nTJ} \rangle$; and T is the number of scenarios shown to a single decision-maker, equal to four in our case.

Eqs. (1) and (8) may be combined and marginalized over classes, to yield the unconditional probability of observing the vectors of choices $\mathbf{y}_n$ and $\mathbf{w}_n$, which in turn may be combined iteratively over decision-makers to yield the following likelihood function for the

Table 4
Summary statistics for LCCMs with varying numbers of classes.

Classes	Parameters	Log-likelihood	$\bar{\rho}^2$	AIC	BIC
2	52	−16,333	0.259	32,770	33,097
3	84	−16,140	0.266	32,448	32,976
4	116	−15,829	0.278	31,889	32,619
5	148	−15,719	0.282	31,733	32,664
6	180	−15,653	0.283	31,666	32,798

data:

$$L(\alpha, \beta, \gamma | y, \mathbf{w}, \mathbf{x}, \mathbf{z}) = \prod_{n=1}^N \sum_{s=1}^S [P(y_n, \mathbf{w}_n | q_{ns} = 1) P(q_{ns} = 1)] \quad (9)$$

The unknown model parameters α , β and γ may be estimated by maximizing the likelihood function. All models for this study were estimated using the software package PythonBiogeme (Bierlaire, 2016). As our sample was stratified exogenously based on demographic and geographic variables, we do not reweight the sample during model estimation (Ben-Akiva and Lerman, 1985).

7. Estimation results

We estimated a number of LCCMs with different model specifications, where we varied the explanatory variables, the functional form of the utilities, and the number of classes. Our dataset comprised 3,985 individuals, each of whom were shown four different choice scenarios. To facilitate comparison, Table 4 enumerates for each model the number of parameters estimated, the log-likelihood at convergence, McFadden's adjusted rho-bar-squared ($\bar{\rho}^2$), the Akaike Information Criterion (AIC), and the Bayesian Information Criterion (BIC). The reader should note that McFadden's adjusted rho-bar-squared and the AIC are equivalent measures of fit. Based both on statistical measures of fit and behavioral interpretation, we select the five-class LCCM as the preferred model specification. In terms of fit, the five-class LCCM has the lowest BIC, and the second lowest AIC after the six-class LCCM. In terms of the signs and relative magnitudes of the different model parameters and the accompanying behavioral interpretation of each of the latent classes, results for the five-class LCCM proved to be the most satisfying.

The final five-class model specification has a McFadden's adjusted R-squared of 0.281. The class membership model included various geographic and demographic characteristics as the explanatory variables $\mathbf{z}_n$, and the corresponding estimates for the model parameters γ_s are enumerated in Table 5. For the purpose of identification, we chose Class 1 as the reference class, i.e. the effect of all variables on the propensity to be in Class 1 is constrained to be zero, and the relative effects of each variable on the corresponding propensity to be in other classes must be evaluated against Class 1. Note that the parameters corresponding to each of the demographic characteristics are significant at the 95% level across at least one of the classes. The parameters corresponding to the geographic characteristics were not found to be statistically significant, but we have retained them in the final specification for the sake of completeness. In some cases, statistical insignificance can still offer valuable policy insights. For example, knowing that MaaS preferences for inner regional residents are not statistically significantly different from MaaS preferences for outer regional residents is valuable, as it suggests that the same approach to MaaS service provision can be applied to both regional contexts. Similar arguments can be made for the other statistically insignificant parameters as well. The class-specific choice models included the availability of different transport modes and services, level of ticketing and booking integration, features such as availability of real-time information and personalization, monthly costs, and subscription model as explanatory variables $\mathbf{x}_n$. Sensitivity to the availability of different transport modes and services was allowed to vary based on whether the subscription model offers pay-as-you-go or unlimited access. Corresponding estimates for the model parameters α_s and β_s are enumerated in Table 6.

The classes have been ordered in terms of their decreasing willingness to use MaaS, and their increasing dependence on the private car. Estimation results for the class membership model and the class-specific choice models provide information on how the classes differ from one another in terms of the kinds of decision-makers that belong to each class (γ_s), the relative importance that they attach to different MaaS service attributes (β_s), and their willingness to purchase access to MaaS schemes (α_s).

To further underscore behavioural differences across classes, we calculate posterior class membership probabilities for each decision-maker in our sample, based on their indicated choices to each of the four scenarios, and reweight them to adjust for geographic and demographic differences between our sample and the Australian population. These weighted probabilities are used to draw out additional differences across the classes, in terms of the geographic, demographic and attitudinal characteristics of the individuals belonging to each class. In particular, we compare the mean proportions for each class against the national average. For example, 30 per cent of individuals belonging to Class 1 are predicted by our model to be between 18 and 29 years, compared to 17 per cent across the national population (of those 18 years or older), leading us to conclude that individuals belonging to Class 1 tend to be younger.

As mentioned before, the classes have been ordered such that the average probability of purchasing MaaS decreases in going from Class 1 to Class 5. These probabilities were calculated as follows: We enumerated all possible MaaS schemes using the attributes and levels defined in Table 1, resulting in $2^{12} = 4096$ schemes in total. For each scheme, we used the discrete choice model to predict the probability that a particular respondent would purchase the scheme if it were available today. These probabilities were averaged

Table 5
Class membership model.

Variable	Class 1 (reference)		Class 2		Class 3		Class 4		Class 5	
	est.	t-stat	est.	t-stat	est.	t-stat	est.	t-stat	est.	t-stat
Class-specific constant	0.000	-	-0.474	-0.79	-0.275	-0.58	1.010	3.14	1.380	4.84
Place of residence										
Inner city	0.000	-	0.576	1.09	0.732	1.85	0.069	0.26	-0.021	-0.09
Inner regional	0.000	-	0.381	0.51	0.653	1.46	0.162	0.49	0.079	0.27
Outer regional (reference)	0.000	-	0.000	-	0.000	-	0.000	-	0.000	-
Age										
30 years or younger	0.000	-	-1.110	-3.30	-0.043	-0.21	-1.590	-7.31	-1.200	-6.47
31 – 64 years old (reference)	0.000	-	0.000	-	0.000	-	0.000	-	0.000	-
65 years or older	0.000	-	-0.184	-0.43	-0.391	-0.80	0.395	1.67	0.671	3.15
Other demographic variables (binary except for weekly household income)										
College educated	0.000	-	-0.174	-0.83	0.025	0.14	-0.406	-2.77	-0.530	-4.17
Employed full time	0.000	-	-0.519	-1.81	-0.676	-2.87	-0.729	-3.60	-0.651	-3.77
Employed part time	0.000	-	-0.354	-1.14	-0.363	-1.50	-0.318	-1.62	-0.475	-2.70
Male	0.000	-	-0.532	-2.29	-0.251	-1.46	-0.553	-3.78	-0.292	-2.30
Have children living at home	0.000	-	-0.499	-2.09	-0.294	-1.66	-0.505	-3.40	-0.504	-3.91
Weekly household income (\$1,000), if disclosed	0.000	-	0.172	1.38	0.216	2.23	0.191	2.35	0.250	3.49
Income not disclosed	0.000	-	0.595	1.51	0.694	2.33	0.733	2.92	1.100	4.93

Table 6
Class-specific choice models.

Variable	Class 1		Class 2		Class 3		Class 4		Class 5	
	est.	t-stat	est.	t-stat	est.	t-stat	est.	t-stat	est.	t-stat
Mean utility of not purchasing MaaS	-0.992	-4.68	0.384	0.63	1.879	7.71	6.136	7.18	6.053	7.98
<i>MaaS service attributes</i>										
Booking integration	0.000	-	0.038	0.12	0.049	0.65	0.433	2.47	0.000	-
Ticketing integration	0.091	1.33	0.076	0.28	0.049	0.57	0.560	2.51	0.060	1.00
Real-time information	0.129	1.97	0.225	0.44	0.166	1.85	0.390	1.84	0.182	3.48
Personalization	0.057	0.88	0.102	0.34	0.082	1.04	0.358	2.02	0.102	1.80
Monthly subscription cost (\$100)	0.000	-	-24.950	-2.17	-0.129	-2.40	-33.172	-9.95	0.000	-
<i>Transport services included – Pay-as-you-go access</i>										
Local public transport	0.488	5.07	1.430	2.63	1.000	7.72	1.350	5.55	0.787	9.59
Long distance public transport	0.493	5.73	0.737	1.99	0.521	4.61	0.738	3.91	0.502	7.07
Taxis	0.349	3.57	0.866	2.11	0.330	2.69	0.949	3.27	0.624	6.28
Rideshare	0.242	2.97	0.886	2.28	0.532	5.19	0.573	3.19	0.330	5.20
Carshare	0.303	3.55	0.525	2.05	0.095	0.89	0.357	1.50	0.261	3.36
Car rental	0.336	4.07	0.209	0.50	0.157	1.07	0.172	1.02	0.554	7.33
Bikeshare	0.097	1.37	0.058	0.29	0.120	1.32	0.000	-	0.000	-
<i>Transport services included – Unlimited access</i>										
Local public transport	0.390	3.56	13.969	2.07	0.666	4.03	12.868	8.19	0.591	6.27
Long distance public transport	0.095	0.59	13.276	1.38	0.355	1.68	84.667	9.92	0.334	2.70
Taxis	0.311	3.16	37.194	1.77	0.426	2.80	29.120	3.07	0.545	6.94
Rideshare	0.181	1.42	22.324	3.77	0.000	-	21.631	19.70	0.000	-
Carshare	0.145	1.24	36.658	1.78	0.360	1.94	28.084	2.97	0.159	1.50
Car rental	0.376	3.02	0.000	-	0.263	1.88	0.000	-	0.385	3.93
Bikeshare	0.002	0.02	0.590	0.23	0.070	0.53	0.000	-	0.000	-

Table 7
High-level summary of different market segments, or classes.

	Class I: MaaS enthusiasts	Class II	Class III	Class IV	Class V: Car dependents
Share of the Australian population	14 per cent	7 per cent	17 per cent	22 per cent	41 per cent
Average MaaS purchase probability	87 per cent	51 per cent	33 per cent	2 per cent	1 per cent
MaaS use	Prefer pay-as-you-go access to local and long distance public transport, taxis and car rentals; likely to use MaaS for all travel	Willing to pay \$130 per month for unlimited access to either taxis or carshare; most likely to use for one-off social trips	Prefer pay-as-you-go access to local and long distance public transport, rideshare and taxis; most likely to use for one-off social trips	Willing to pay \$230 per month for unlimited access to long-distance public transport; most likely to use for one-off social trips	Unwilling to use MaaS, regardless of service attributes
Attitudes towards MaaS and car ownership	MaaS could help reduce car dependence and car ownership				MaaS unlikely to have effect on car dependence or car ownership
Geography	Evenly spread across metro, regional and remote areas				More likely to live in regional and remote areas
Demography	More likely to be younger, male, college educated, employed, have children at home	More likely to be middle aged, female, college educated, high household income	More likely to be young, college educated, single and living with parents, high household income	More likely to be old, female, not college educated, retired, empty nesters	More likely to be old, not college educated, retired, empty nesters
Current travel behaviour and attitudes	High overall travel needs, high motorcycle ownership, high use of mobility devices				Low opinion and infrequent use of public transport and carsharing
Average self-reported travel costs	\$185 per capita per week	\$121 per capita per week	\$136 per capita per week	\$98 per capita per week	\$107 per capita per week

over all MaaS schemes, and reweighted to adjust for differences between our sample and the Australian population, to arrive at the mean predicted probability that an Australian belonging to the class or market segment would purchase MaaS if it were available today.

We additionally characterize the classes in terms of differences in MaaS use, attitudes towards MaaS and car ownership, and current travel behaviour and attitudes. Differences in MaaS use have been inferred from the class-specific choice models reported in Table 6. For example, class-specific willingness-to-pay measures for unlimited access to a particular mode have been computed by taking the ratio of the corresponding coefficient for that mode and the price coefficient from the class-specific choice model. Recall from Section 4 that for each MaaS scenario, respondents were asked to indicate the kinds of trips they would make using their preferred MaaS scheme. Recall further that respondents were asked about their attitudes towards MaaS and car ownership using five Likert-scale statements, reported previously in Fig. 4. They were also asked about their current use of different transport modes, and their opinions about private car ownership, public transport and carsharing, using additional Likert-scale statements. This information has been incorporated in our post-estimation analysis to further delineate differences across the classes.

Over subsequent paragraphs, we describe in greater detail each of the five classes identified by the model. These descriptions are summarized in Table 7.

Class 1: Comprising 14 per cent of the national population, individuals belonging to this class have the greatest likelihood of purchasing MaaS, with a very high average purchase probability of 87 per cent across all possible MaaS schemes tested by our analysis. Local public transport, long distance public transport, taxis and car rental services are the most popular modes for these individuals, and bikeshare is the least popular. Expected use of MaaS is evenly spread across all trip purposes. Individuals belonging to this class are more likely to be younger (30 per cent are under 30 years old), male (55 per cent), college educated (50 per cent), employed (73 per cent), and have children at home (42 per cent). In terms of current travel behaviour, those that travel frequently, have high rates of motorcycle ownership, are dependent on mobility devices, and/or have high perceived travel costs, are most likely to belong to this class.

Class 2: Comprising 7 per cent of the national population, individuals belonging to this class have a high average purchase probability of 51 per cent across all possible MaaS schemes tested by our analysis. Taxis and carshare services are the most popular modes of transport: our model predicts that nearly all individuals belonging to the class would purchase a MaaS scheme that offers unlimited access to either mode for under \$130, and both modes for a combined \$260. Expected use of MaaS is highest for one-off social trips, and lowest for commuting trips. Like Class 1, these individuals are more likely to be college educated (44 per cent) and employed (63 per cent). Unlike Class 1, they are more likely to be middle aged (74 per cent are between 30 and 65 years old) and female (57 per cent), and have high average household incomes (\$1606 per week).

Class 3: Comprising 17 per cent of the national population, individuals belonging to this class have a moderate average purchase probability of 33 per cent across all possible MaaS schemes tested by our analysis. Local public transport is the most popular mode of transport, followed by rideshare, long distance public transport, and taxis. Individuals belonging to this class exhibit a strong preference for pay-as-you-go access. Like Class 1, they are more likely to be younger (32 per cent are under 30 years old) and college educated (49 per cent). Like Class 2, they are more likely to have high average household incomes (\$1664 per week). Across all classes, Class 3 has the highest proportion of single adults living with their parents (14 per cent, compared to the national average of 8 per cent).

Class 4: Comprising 22 per cent of the national population, individuals belonging to this class have a near zero average purchase probability of 2 per cent across all possible MaaS schemes tested by our analysis. However, they have a strong preference for unlimited access to long distance public transport: our model predicts that nearly all individuals belonging to the class would purchase a MaaS scheme that offers unlimited access to that mode for under \$230. Expected use of MaaS is primarily limited to one-off social trips. Compared to previous classes, individuals belonging to Class 4 are more likely to be older (average age of 51 years; 22 per cent over 65 years old), retired from the work force (24 per cent, compared to 8 per cent for Class 1) and living in regional and remote areas (22 per cent, compared to 14 per cent for Classes 2 and 3), and less likely to be college educated (35 per cent).

Class 5: Comprising the remaining 41 of the national population, like Class 4, individuals belonging to this class have a near zero average purchase probability of 1 per cent across all possible MaaS schemes tested by our analysis. However, unlike Class 4, they appear resolute in their unwillingness to purchase MaaS, regardless of service attributes. In terms of geographic and demographic characteristics, individuals belonging to Class 4 and Class 5 have much in common with regards to age, education, employment status, and place of residence. In terms of distinguishing characteristics, individuals belonging to Class 5 are more likely to be male (51 per cent, compared to 44 per cent for Class 4) and older (27 per cent are over 65 years old).

These findings reveal multiple segments in the market that different MaaS systems could target, in an attempt to maximize patronage and uptake. For example, MaaS systems that offer pay-as-you-go access to local and long distance public transport, taxis, and rideshare services are most likely to appeal to consumers belonging to Classes 1 and 3. Similarly, MaaS systems that offer unlimited access to taxis and carshare services are most likely to appeal to Class 2. And finally, MaaS systems that offer unlimited access to long-distance public transport are most likely to appeal to Class 4.

There are also general demographic trends across classes that are worth pointing out. For example, willingness to use MaaS is strongly correlated with age and life cycle stage. Young and middle-aged individuals who are either single or married, with or without children at home, are far more likely to use MaaS. In contrast, older individuals whose children have left home are most unwilling to use MaaS. Education and employment are strongly correlated with willingness to use MaaS as well, with more educated and employed individuals being more likely to use MaaS.

Current patterns of travel behaviour and attitudes towards existing transport modes and services serve as excellent indicators of willingness to use MaaS. Individuals unwilling to use MaaS have lower assessments of public transport services in their local

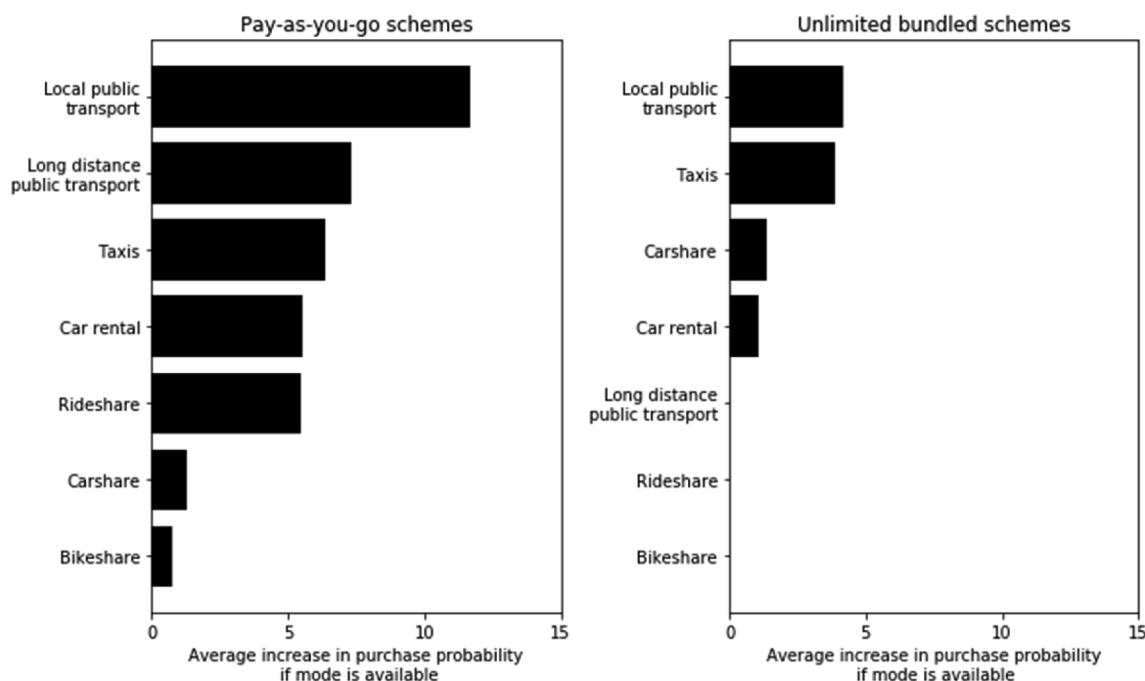


Fig. 5. Average demand elasticities for pay-as-you-go and unlimited bundled MaaS schemes, as a function of access to different transport modes.

neighbourhoods, are generally not open to the idea of carsharing, and are more likely to report that private car ownership is a necessity where they live. And the converse is true for individuals most willing to use MaaS.

Relatedly, the higher an individual's perceived travel costs, the more likely they are to use MaaS. Separately, we also found that most individuals in our sample had underestimated their weekly travel expenditures, in particular the costs of private car ownership. The two findings together suggest that making consumers more aware of the marginal costs of private car ownership and use might be an important mechanism for increasing MaaS' appeal.

8. Policy analysis

We examine some of the implications of our findings for transport practitioners and policy-makers interested in the provision of MaaS. On average, our model finds that consumers prefer pay-as-you-go schemes twice as much as prepaid schemes that offered unlimited access. This is in contrast to findings reported by [Ho et al. \(2018\)](#) and [Matyas and Kamargianni \(2019\)](#); both studies found that consumers in their sample preferred bundled prepaid offerings to pay-as-you-go schemes. However, both studies used convenience sampling to recruit study participants, and the preferences of their sample may not be reflective of the preferences of the general population ([Yeager et al., 2011](#)).

Fig. 5 illustrates consumer preferences for access to different transport modes, as a function of the subscription model, in terms of average demand elasticities as estimated by the model reported in [Section 7](#). For example, on average and all else being equal, pay-as-you-go schemes that offer access to local public transport services are 12 per cent more likely to be purchased than pay-as-you-go schemes that do not offer access to the same. For pay-as-you-go schemes, local public transport is by far the most popular mode, followed by long-distance public transport (i.e. regional bus and train services), taxis, car rentals and rideshare services. Carshare and bikeshares services have limited appeal. For prepaid schemes that offer unlimited access, local public transport and taxis are the only two modes that are strongly preferred. These findings indicate the inclusion of which transport modes is most critical to the adoption of MaaS schemes among Australian consumers. In particular, the importance of local public transport is consistent with findings from both [Ho et al. \(2018\)](#) and [Matyas and Kamargianni \(2019\)](#).

[Table 8](#) enumerates adoption rates of different MaaS schemes among Australian consumers, as predicted by our model. For the sake of simplicity, all our scenarios assume that the MaaS service offers full planning, ticketing and booking integration; and access to both real time information and personalization features. By and large, our model indicates that there is definitely a market for MaaS in Australia. MaaS schemes that offer pay-as-you-go access to transport modes have a predicted adoption rate of between 30 and 46 per cent, depending on the transport modes that they offer access to and their monthly subscription costs. Even prepaid schemes that offer unlimited access to local public transport and taxi services for a high monthly cost of \$500 have a high predicted adoption rate of 18 per cent. Overall, these findings should serve to encourage industry and government players interested in the provision of MaaS in Australia.

Next, we examine how consumer preferences for MaaS vary by state and region. In particular, we compare predicted adoption rates across states and regions for two MaaS schemes. The first scheme offers pay-as-you-go access to all modes; has no monthly

Table 8

Predicted adoption of different MaaS schemes.

MaaS scheme (all schemes assumed to have full planning, ticketing and booking integration; real time information; and personalization features)		Predicted share of Australian population that would purchase scheme
Pay-as-you-go access to all modes	No monthly subscription costs	45.9%
	\$5 monthly subscription	39.5%
	\$10 monthly subscription	37.0%
Pay-as-you-go access to local public transport, long distance public transport and taxis	No monthly subscription costs	35.8%
	\$5 monthly subscription	31.8%
	\$10 monthly subscription	29.4%
Unlimited access to local public transport and taxis	\$500 monthly subscription	18.1%
Unlimited access to local public transport	\$150 monthly subscription	17.4%

Table 9

Predicted proportion of residents living in metro, regional and remote areas across different states and territories that would purchase a MaaS scheme that offers pay-as-you-go access to all transport modes, has no monthly subscription costs, offers full planning, ticketing and booking integration, real time information, and personalization features.

State/Territory	Total	Metro	Regional and Remote
New South Wales	48.0%	47.5%	49.3%
Victoria	44.7%	49.2%	36.3%
Queensland	44.0%	45.8%	41.7%
South Australia	39.3%	45.4%	28.4%
Western Australia	41.5%	42.2%	39.5%
Northern Territory	NA	NA	NA
Tasmania	NA	NA	NA
Australian Capital Territory	49.0%	49.0%	NA

Table 10

Predicted proportion of residents living in metro, regional and remote areas across different states and territories that would purchase a MaaS scheme that offers unlimited access to local public transport and taxi services, at a monthly cost of \$500, offers full planning, ticketing and booking integration, real time information, and personalization features.

State/Territory	Total	Metro	Regional and Remote
New South Wales	18.7%	18.1%	20.2%
Victoria	16.6%	19.7%	10.8%
Queensland	16.9%	16.9%	16.9%
South Australia	13.7%	16.0%	9.4%
Western Australia	15.4%	15.6%	14.7%
Northern Territory	NA	NA	NA
Tasmania	NA	NA	NA
Australian Capital Territory	18.7%	18.7%	NA

subscription costs; has have full planning, ticketing and booking integration; and offers real time information and personalization features. The second scheme offers unlimited access to local public transport and taxi services; has a \$500 monthly subscription cost; has have full planning, ticketing and booking integration; and offers real time information and personalization features.

Recall per [Table 8](#) that the first scheme has a national predicted adoption rate of 45.9 per cent, and the second scheme has a national predicted adoption rate of 18.1 per cent. [Tables 9 and 10](#) show how these numbers break down across the states and territories. On aggregate, New South Wales, Victoria, Queensland and the Australian Capital Territory have bigger potential markets than Western Australia and South Australia. Across metropolitan regions, good markets for MaaS pilots would be Melbourne, Canberra and Sydney. However, by and large, the potential market for MaaS in metropolitan regions appears to be sizable across all states, especially for pay-as-you-go schemes.

Surprisingly, our findings indicate that regional and remote New South Wales has a bigger potential market than most metropolitan regions. We examine this finding in more detail through a plot of the predicted proportion of New South Wales residents across different regions that would purchase different MaaS schemes, shown in [Fig. 6](#). In drawing up this map, and the map that follows, we followed the same procedure described in [Section 7](#) to calculate the average MaaS purchase probabilities by class membership. We enumerated all possible MaaS schemes, calculated purchase probabilities for each scheme for each respondent in our sample, reweighted the probabilities to adjust for differences between our sample and the Australian population, and averaged them over particular geographic areas, to arrive at the mean predicted probability that an Australian living in that geographic area would purchase MaaS if it were available today.

The reader should note that the following regional and remote communities in particular have high potential demand for MaaS:

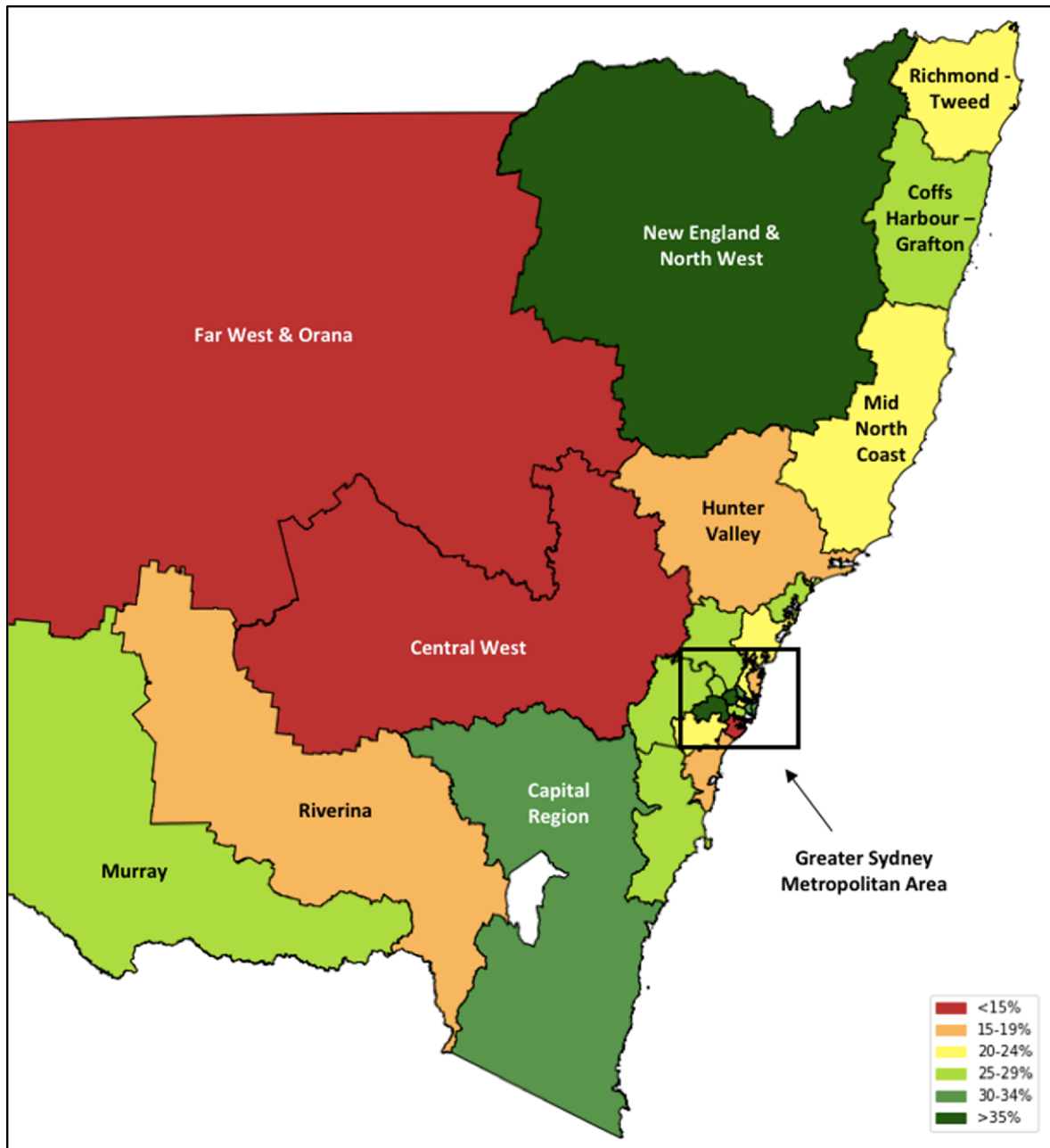


Fig. 6. Predicted proportion of New South Wales residents across different regions that would purchase MaaS if it were available today.

regional communities along the New South Wales coastline, both north and south of Sydney; regional communities in the far west suburban and exurban regions surrounding Sydney; agricultural communities in the Murray basin; and mining communities in New England and North West New South Wales. While the populations in these regions might not be large enough to commercially support the launch of MaaS pilots, our analysis indicates that residents in these regions are more willing than the average Australian to embrace this new paradigm of transport service provision.

As an illustration of how preferences for MaaS might vary within the same metropolitan region, Fig. 7 plots the predicted proportion of residents in the Greater Sydney metropolitan area that would purchase MaaS. Expectedly, residents in richer car-dependent neighborhoods, such as Northern Beaches and Sutherland Shire, have the lowest potential demand for MaaS. Residents in inner city neighborhoods have high potential demand for MaaS. However, and perhaps surprisingly, the greatest potential demand for MaaS appears not to be in the inner-city neighborhoods, but in the suburbs surrounding Sydney, such as Parramatta and South West Sydney, which have lower average household incomes and depend more heavily on the public transport network, in particular the rail network, to fulfill their mobility needs.

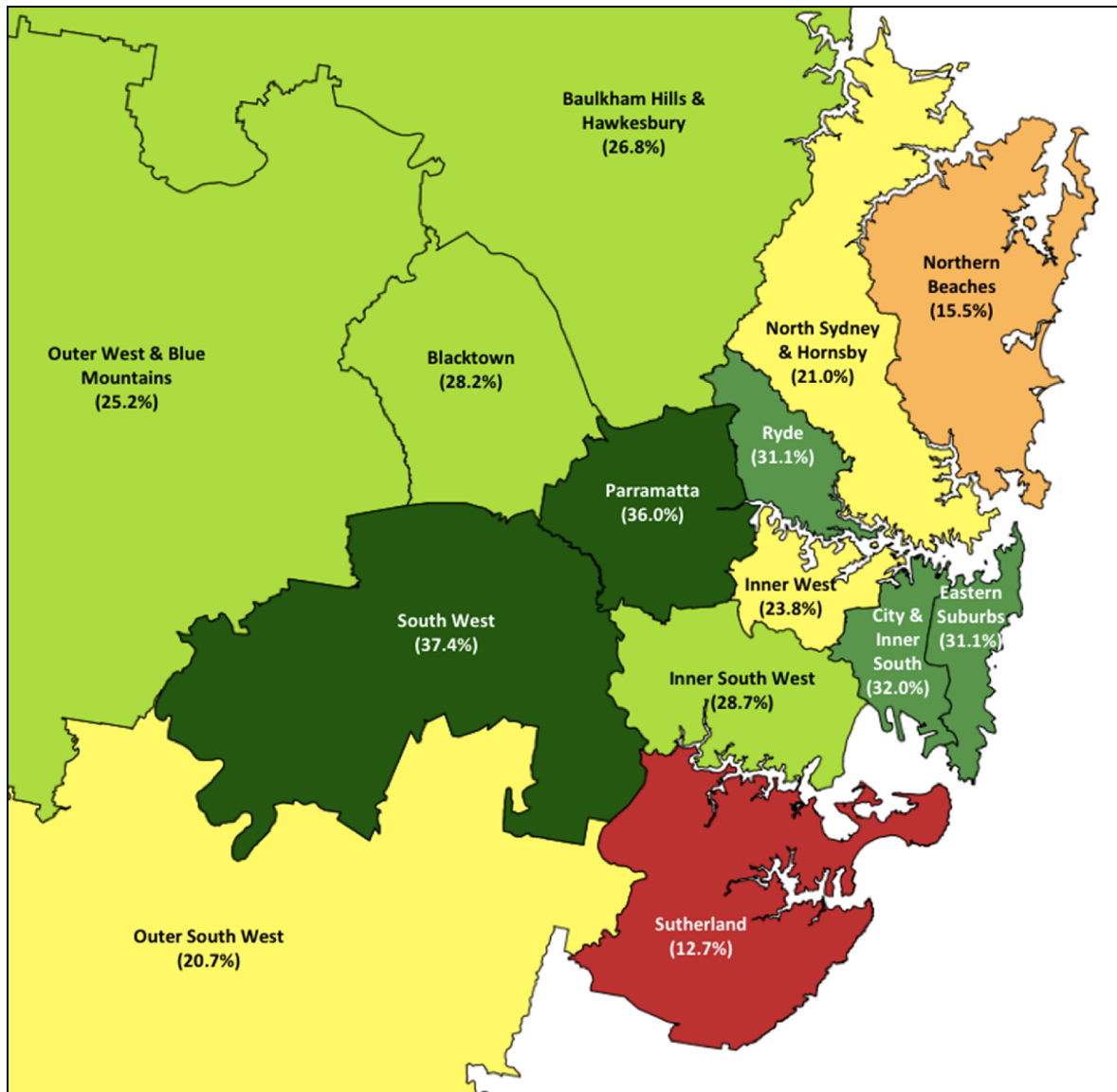


Fig. 7. Predicted proportion of Greater Sydney Metropolitan Area residents across different neighborhoods that would purchase MaaS if it were available today.

The reader should note however that predicted adoption rates specific to a neighborhood or region are subject to greater sampling variance than predicted adoption rates for the entire state, due simply to smaller sample sizes. For example, across neighborhoods and regions shown in Figs. 6 and 7, the sample size varies between 8 and 84 respondents. As such, our findings are indicative of potential demand for MaaS across much of the Greater Sydney metropolitan area and parts of regional New South Wales. However, more data is needed to conclude definitively which neighborhoods and regions in particular should be selected for early trials.

9. Conclusions

The concept of MaaS, or Mobility-as-a-Service, was first introduced in 2014 in Helsinki, Finland as a sustainable, convenient and cost-effective alternative to private car ownership. MaaS offers potential consumers access to multiple transport modes and services, owned and operated by different mobility service providers, through an integrated digital platform for planning, booking and payment. Since its introduction, several different MaaS schemes have been launched across other parts of Europe, such as Mon Chauffeur in France, Mobil-Flat in Germany and Renfe-as-a-Service in Spain, and interest from transport practitioners and policy-makers worldwide indicates that similar schemes will soon become commercially available in other parts of the world as well (e.g. Abdullah, 2019; iMOVE, 2019).

This study examined consumer preferences for MaaS in Australia. We surveyed 3985 geographically and demographically

representative Australians nationwide. Our analysis confirmed that there is definitely a market for MaaS in Australia. For example, we found that 32 per cent of the Australian population would adopt a MaaS scheme that offers pay-as-you-go access to local public transport, long distance public transport and taxi services for a monthly charge of \$5 if such a service were available in the market today. On average, consumers prefer pay-as-you-go schemes to bundled schemes that offer unlimited access to one or more transport modes and services at fixed monthly costs. Local public transport, taxis and long-distance public transport are the most popular transport services; bikeshare is the least popular. Willingness to use MaaS is strongly correlated with age and lifecycle stage: young individuals who are employed full-time are most likely to use MaaS; older adults who have retired from the workforce and whose children have left home are least likely to use MaaS. Given both the latent consumer demand for MaaS, and the interest shown by industry and government in delivering such a service, it is likely only a matter of time before MaaS becomes commercially available in Australia. Our analysis identifies Melbourne, Canberra and Sydney as good markets for MaaS trials and early launches.

CRediT authorship contribution statement

Akshay Vij: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing - original draft, Writing - review & editing. **Stacey Ryan:** Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Writing - original draft. **Spring Sampson:** Data curation, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Writing - original draft. **Susan Harris:** Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision.

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